

UDAY KIRAN REDDY K

Machine Learning Engineer

✉ udayreddy1642@gmail.com ☎ (839) 282-2232 📍 USA 🔗 [linkedin.com/in/udayreddy1642/](https://www.linkedin.com/in/udayreddy1642/)

Machine Learning Engineer

Machine Learning Engineer with proven experience designing and deploying scalable AI solutions in production environments. Proficient in building end-to-end machine learning pipelines using Python, PyTorch, TensorFlow, and Azure ML. Expert in optimizing data workflows to reduce operational costs while enhancing predictive accuracy. Strong technical background in full-stack integration (Angular, Node.js) and cloud services, bridging the gap between complex models and actionable business insights.

PROFESSIONAL EXPERIENCE

Software Engineer Intern, Publicis Sapient	01/2024 – 07/2024 Bengaluru, India (Hybrid)
• Collaborated with DTCC on post-trade systems processing \$2T+ in daily transactions, ensuring MiFID II compliance for reconciliation and reporting pipelines. • Engineered Python + Azure Functions automation and integrated RabbitMQ to orchestrate trade reconciliation, automating 12 manual jobs and reducing reconciliation effort by 40% (~500 hours saved annually). • Built and maintained Grafana dashboards backed by Azure Data Explorer and Cosmos DB for real-time health monitoring, cutting mean time to detection/escalation by 50%. • Led cross-functional incident response for Oracle DB and SQL Server, preventing 2 production incidents and preserving data integrity for 25 downstream reports/dashboards.	
Software Developer Intern, ArkaShine Innovations	09/2023 – 10/2023 Bengaluru, India (Hybrid)
• Designed and deployed PyTorch models for precision irrigation using IoT sensor + weather data, increasing scheduling accuracy by 30% and reducing water usage in pilot farms by 25%. • Optimized inference pipeline (preprocessing + hyperparameter tuning), achieving 3x throughput and reducing latency by 50%. • Contributed to pilot deployments that delivered estimated annual savings of ~\$12,000 for medium-sized farms via reduced water waste and operational efficiency.	

EDUCATION

M.S. in Computer Science (Data Science), University of North Carolina at Charlotte GPA: 3.77/4.00	05/2026
B.Tech in Computer Science, M. S. Ramaiah University of Applied Sciences, India GPA: 8.91/10.00	2020 – 2024

AREAS OF EXPERTISE

- MLOps, ETL/ELT Pipeline Development, Azure Data Services (Cosmos DB, ADX), Docker, SQL, Real-time & Batch- parallel Processing, Python, PyTorch, TensorFlow, Database, Design Performance, Tuning & Optimization. Linux
- Full-Stack Development (Angular/Node.js) Scikit-learn Pandas Data Visualization (Tableau/Grafana) Cross-functional Leadership Change Management
- Git DigitalOcean AWS Governance & Secured Access Control Machine Learning & Deep Learning REST APIs

PUBLICATIONS

Springer Nature (SmartCom 2024)

- First Author of a research paper titled "Machine Learning-Powered Car Recommendation System: A Content-Based and Collaborative Approach."
- Developed a hybrid recommender engine using Random Forest and collaborative filtering that outperformed baseline Neural Networks by 18% in precision.
- Published in Lecture Notes in Networks and Systems (DOI: 10.1007/978-981-97-1326-4_28).

PROJECTS

Nest-Sense-AI (GenAI/RAG), LLAMA, LangChain, XGBoost, Vector DB

- Designed a Retrieval-Augmented Generation (RAG) pipeline to query unstructured data with high context accuracy using LLaMA and Vector Databases.
- Executed XGBoost for auxiliary classification tasks and optimized data retrieval workflows using Pandas and Joblib.

Pneumonia Detection, PyTorch, ResNet18, Torchvision

- Built a medical image classifier using Transfer Learning (ResNet18) on chest X-ray datasets to detect pneumonia with high sensitivity.
- Utilized Torchvision for advanced image augmentation and transformation, significantly improving model robustness and generalization.

Full-Stack Analytics Dashboard, Angular 18, Node.js, MySQL, DigitalOcean

- Developed a secure Single Page Application (SPA) using Angular 18 and Node.js with JWT authentication to protect API endpoints.
- Deployed the solution on DigitalOcean (Ubuntu/NGINX) using PM2 for process management and D3.js for real-time data visualization.

Car-Vision-Blackwell, YOLO, OpenCV, Ultralytics

- Implemented a real-time vehicle detection system using YOLOv8 and OpenCV, capable of identifying multiple vehicle classes in dynamic environments.